

SOF NSO Science — Topic Deep-Dive: Earth & Universe

For Ambar, Class III CBSE

1. Sun, Moon, stars and planets

When we look at the sky we see many things.

Object	What it is
Sun	A star — it gives us heat and light in the day
Moon	The Earth's natural satellite — seen at night
Stars	Huge balls of burning gas that make their own light; look tiny because they are very far away
Planets	Big round bodies that go around the Sun; they do not make their own light

The **Sun is a star** — it looks the biggest and brightest only because it is the **nearest** star to the Earth. Stars look like tiny points because they are **very, very far** away.

2. The planets

Eight planets move around the Sun. - **Mercury** is the planet **closest** to the Sun. - **Jupiter** is the **largest** planet. - **Earth** is the planet we live on.

The order of the first few planets from the Sun is: **Mercury, Venus, Earth, Mars...** (Mercury is first, then Venus, then our Earth).

3. Day and night — the Earth's rotation

The Earth **spins** around like a top. This spinning is called **rotation**. - The Earth takes **one day** to spin once. - The side facing the Sun has **day**; the side turned away has **night**. - So when it is day in India, it is night on the other side of the world.

The rotation (spinning) of the Earth gives us day and night.

4. The year — the Earth's revolution

The Earth also moves in a big path **around the Sun**. This is called **revolution**. - The Earth takes about **one year** to go once around the Sun.

Rotation → day and night. Revolution → one year. (Do not mix these two up!)

5. Phases of the Moon

The Moon does **not** make its own light. It shines because it **reflects the Sun's light**.

As the month passes, we see different lit parts of the Moon, so it looks like it changes shape — full moon, half moon, thin crescent. These changing shapes are the **phases of the Moon**.

6. Natural and artificial satellites

A **satellite** is a body that goes around a planet.

Type	Meaning	Example
Natural satellite	Made by nature	The Moon (goes around Earth)
Artificial satellite	Made by people and sent into space	Weather satellites, TV satellites

Artificial satellites help us with **weather news, telephone and TV signals**.

Common traps

The trap	The truth
"The Sun is a planet"	The Sun is a star
"The Moon makes its own light"	The Moon only reflects the Sun's light
"Rotation gives us a year"	Rotation gives day and night; revolution gives a year
"Earth is the largest planet"	Jupiter is the largest; Earth is where we live
"The Moon is an artificial satellite"	The Moon is a natural satellite
"Stars are small"	Stars are huge ; they only look tiny because they are far

High-frequency question types

1. **What is it:** "The Sun is a ____" → star.
 2. **Planet facts:** "Which planet is closest to the Sun?" → Mercury; "largest?" → Jupiter.
 3. **Rotation:** "The spinning of the Earth gives us ____" → day and night.
 4. **Revolution:** "One round of Earth around the Sun = ____" → one year.
 5. **Satellite:** "The Moon is Earth's natural ____" → satellite.
 6. **Moon light:** "The Moon shines because it ____" → reflects the Sun's light.
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Key facts checklist

- ☐ The Sun is a star; it gives heat and light.
 - ☐ The Moon is Earth's natural satellite; seen at night.
 - ☐ Stars make their own light; look tiny because they are far.
 - ☐ Eight planets go around the Sun.
 - ☐ Mercury = closest to the Sun; Jupiter = largest planet.
 - ☐ Order from Sun: Mercury, Venus, Earth, Mars...
 - ☐ Rotation (spinning) → day and night.
 - ☐ Revolution (round the Sun) → one year.
 - ☐ Moon reflects the Sun's light; phases = changing shapes.
 - ☐ Natural satellite = Moon; artificial satellite = made by people.
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Quick drill (10 questions)

1. The Sun is a star or a planet?
2. Which planet is closest to the Sun?
3. Which is the largest planet?
4. How many planets go around the Sun?
5. On which planet do we live?

6. The spinning of the Earth gives us ____.
7. One round of the Earth around the Sun gives one ____.
8. The Moon is Earth's natural ____.
9. Does the Moon make its own light?
10. A satellite made by people is called a(n) ____ satellite.

Answers: 1. Star 2. Mercury 3. Jupiter 4. Eight 5. Earth 6. Day and night 7. Year 8. Satellite 9. No — it reflects the Sun's light 10. Artificial